ROBAYET ROWSHAN SIAM

Gazipur, Dhaka, Bangladesh | +880 1323128673 | rrsiam6@gmail.com | GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate at United International University (CGPA 3.82/4.00) with hands-on experience in AI-powered applications, algorithms and software systems. Proficient in Python, Java and C/C++, with practical work in machine learning, deep learning, reinforcement learning and explainable AI. Seeking research and software engineering roles on intelligent, technology-driven solutions.

EDUCATION

BSc in Computer Science and Engineering 2023 – Present
United International University, Dhaka, Bangladesh
CGPA: 3.82/4.00 Expected Graduation: February 2027

RELEVANT COURSEWORK

Core & Architecture: Data Structures & Algorithms, System Design & Analysis, Software Engineering
Web & Interaction: Web Programming, Human-Computer Interaction
Data & Intelligent Systems: Database Management Systems, Artificial Intelligence, Machine Learning, Data Mining, Digital Image Processing

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, PHP
Web Technologies: HTML, CSS, JavaScript
Database: MySQL
AI Libraries & Frameworks: Scikit-learn, Librosa, Whisper, DeepFace, NumPy, Pandas, Matplotlib
Development Tools: Git, GitHub, VS Code, Kaggle, Google Colab, Overleaf

ACADEMIC PROJECTS

AI Confidence Coach – Multimodal ML System for Confidence Measure *GitHub*
Python, Scikit-learn, Librosa, Whisper, SHAP, Stable-Baselines3 (PPO), DeepFace, FastAPI

- Predicts speaking confidence (0–100) from 34 audio features (Librosa, Whisper) fused with DeepFace facial emotion scores, and explains each score in plain language using SHAP.
- Built a 550-clip dataset from RAVDESS, CREMA-D and TED-LIUM rated by 3 annotators (ICC = 0.737); trained a Random Forest regressor (MAE = 7.53 on a 0–100 scale).
- SHAP revealed pause ratio, speech rate and pitch variation as key drivers, and showed Whisper strips filler words, making the filler feature unreliable.
- Designed a PPO agent selecting among 8 coaching exercises; evaluated on 50 held-out clips and real voice recordings, diagnosing score compression and RL policy collapse.

MeramotHub – Local Service Booking Platform *Live Demo / GitHub-Frontend / GitHub-Backend*
React, Node.js, Supabase (PostgreSQL), JWT

- Built a four-role platform (Customer, Service Provider, Area Manager, Admin) with role-based dashboards, location-based worker search, ratings and job tracking.
- Designed an escrow payment flow (hold, completion confirmation, commission deduction, release) with MFS payments.
- Implemented complaint handling that freezes payment and assigns an Area Manager, plus a Groq-powered AI chatbot for recommendations.

Fine-Tuning TigerLLM-1B for Bengali Medical Question Answering *Live Demo / Model / GitHub*
Python, PyTorch, Hugging Face Transformers, PEFT, TRL, QLoRA, Gradio

- Fine-tuned TigerLLM-1B-it for Bengali medical question answering with QLoRA on a single Tesla T4, training only 13M parameters (≈1.29% of the model) in under an hour.
- Built the end-to-end pipeline over 901 Bengali medical QA pairs (810/91 split) with a chat prompt template and chain-of-thought answers; 4-, 8- and 12-epoch sweeps showed step 200 (≈4 epochs) optimal before validation loss began to rise.
- Merged the best checkpoint to Hugging Face Hub and deployed a Gradio app with adjustable token length and temperature plus pre-loaded Bengali medical queries.
- Improved over the base model on sample diagnoses: accurate Bengali clinical terminology and complete, structured answers instead of hallucinated or truncated output.